

Lemma 3.1.4. *The block graph of a connected graph is a tree.* $\square$

Lemma 3.1.4 generalizes to graphs of higher connectivity: every $(k-1)$ -connected graph has a canonical tree-like decomposition that separates all its ‘ k -blocks’. See Theorem 12.3.7 for the precise statement, and Exercise 17 in Chapter 12 for the case of $k = 3$.

3.2 The structure of 3-connected graphs

In this section we describe how every 3-connected graph can be obtained from a K^4 by a succession of elementary operations preserving 3-connectedness. We then prove a theorem of Tutte about the algebraic structure of the cycle space of 3-connected graphs; this will play an important role again in Chapter 4.5.

Proposition 3.1.1 describes how the 2-connected graphs can be constructed inductively, starting from a cycle. All the graphs constructed in the process were themselves 2-connected, so the graphs constructible in this way are precisely the 2-connected graphs. We shall now do something similar for 3-connected graphs. We shall prove that every 3-connected graph $G \neq K^4$ can be turned into a smaller 3-connected graph in two ways: by deleting an edge (and suppressing any vertices of degree 2 that may arise), and by contracting an edge. Inverting these processes will give us two independent ways of building all 3-connected graphs from a K^4 .

$G \div e$ Given an edge e in a graph G , we write $G \div e$ for the *multigraph* obtained from $G - e$ by suppressing any end of e that has degree 2 in $G - e$.³

Lemma 3.2.1. *Let e be an edge in a graph G . If $G \div e$ is 3-connected, then so is G .*

(1.4.2) *Proof.* Thinking of G as obtained from $G \div e$ by adding e , let us call the vertices of $G \div e$ the *old* vertices of G , and any other vertex of G (which will be an end of e) a *new* vertex. Remembering that $G \div e$, being 3-connected, has no parallel edges, it is easy to see that, in G , no two vertices x_1, x_2 can separate a new vertex from all the old vertices. So it suffices to show that $\{x_1, x_2\}$ cannot separate two old vertices. If they did, then those old vertices would be separated in $G \div e$ by x'_1 and x'_2 , where either $x'_i = x_i$ or, if x_i is new, x'_i is the edge of $G \div e$ subdivided by x_i . By Proposition 1.4.2, this contradicts our assumption that $G \div e$ is 3-connected. $\square$

³ See Chapter 1.10 for the formal definition of suppressing vertices in a multigraph. Recall also that 3-connected multigraphs cannot have multiple edges. Since parallel edges arising when a vertex is suppressed are not deleted, our assumption in Lemma 3.2.1 that the multigraph $G \div e$ is 3-connected implies that no parallel edges arise when it is formed from the graph G . Thus $G \div e$, too, is in fact a graph.

Lemma 3.2.2. *Every 3-connected graph $G \neq K^4$ has an edge e such that $G \div e$ is another 3-connected graph.*

Proof. We start by showing that G contains a TK^4 . Let C be a shortest cycle and $P = u \dots v$ a C -path in G . Then $\dot{P} \neq \emptyset$ since C is induced, so $G - \{u, v\}$ contains a C - P path Q . Now $C \cup P \cup Q = TK^4$.

Thus, $TK^4 \subseteq G \neq K^4$; choose $H = TJ \subseteq G$ with some 3-connected $J \neq G$ so that $\|H\|$ is maximum. We shall find an edge e in G such that $G \div e = J$. H

As G is 3-connected but proper subdivisions of J are not, we have $H \neq G$. Let $P = u \dots v$ be an H -path in G , chosen if possible so that

$$u \text{ and } v \text{ do not lie on the same (subdivided) edge of } J. \quad (*) \quad P = u \dots v$$

If P violates $(*)$ then $uv \in J = H$. Indeed, since G is 3-connected, the vertices subdividing an edge xy of J could be joined by an H -path P' in $G - \{x, y\}$ to a vertex not on that subdivided edge; as P' would satisfy $(*)$, this would contradict our choice of P . So $H = J$. Our assumption that P violates $(*)$ now means that $uv \in J$. Since G has no parallel u - v edges, P has an inner vertex. Now $(H - uv) \cup P$ is another TJ with more edges than H , contradicting our choice of H .

Therefore P satisfies $(*)$. Suppressing any vertices of degree 2 in $H \cup P$ we obtain a multigraph J' such that $J' \div e = J$, where e is the edge corresponding to P . By $(*)$ the edge e is not parallel to an edge of J , so J' , like J , is in fact a graph. By Lemma 3.2.1, J' is 3-connected. As $TJ' = H \cup P \subseteq G$ has more edges than H , our choice of J with $\|H\|$ maximum implies $J' = G$. Thus, $G \div e = J$ as claimed. $\square$

Theorem 3.2.3. (Tutte 1966)

A graph G is 3-connected if and only if there exists a sequence $G_0, \dots, G_n$ of graphs such that

- (i) $G_0 = K^4$ and $G_n = G$;
- (ii) G_{i+1} has an edge e such that $G_i = G_{i+1} \div e$, for every $i < n$.

Moreover, the graphs in any such sequence are all 3-connected.

Proof. If G is 3-connected, use Lemma 3.2.2 to find $G_n, \dots, G_0$ in turn. Conversely, if $G_0, \dots, G_n$ is any sequence of graphs satisfying (i) and (ii), then all these graphs, and in particular $G = G_n$, are 3-connected by Lemma 3.2.1. $\square$

Theorem 3.2.3 enables us to construct, recursively, the entire class of 3-connected graphs. Starting from K^4 , we simply add to every graph already constructed a new edge in every way compatible with (ii): between two already existing vertices, between newly inserted subdividing vertices (not on the same edge), or between one old vertex and one new subdividing vertex.

We now turn to our second method of reducing 3-connected graphs to K^4 , by contracting edges. In what follows we only consider graphs, not multigraphs.

[4.4.3] **Lemma 3.2.4.** *Every 3-connected graph $G \neq K^4$ has an edge e such that G/e is again 3-connected.*

Proof. Suppose there is no such edge e . Then, for every edge $xy \in G$, the graph G/xy contains a separator S of at most 2 vertices. Since $\kappa(G) \geq 3$, the contracted vertex v_{xy} of G/xy (see Chapter 1.7) lies in S and $|S| = 2$, i.e. G has a vertex $z \notin \{x, y\}$ such that $\{v_{xy}, z\}$ separates G/xy . Then any two vertices separated by $\{v_{xy}, z\}$ in G/xy are separated in G by $T := \{x, y, z\}$. Since no proper subset of T separates G , every vertex in T has a neighbour in every component C of $G - T$.

We choose the edge xy , the vertex z , and the component C so that $|C|$ is as small as possible, and pick a neighbour v of z in C (Fig. 3.2.1). By assumption, G/zv is again not 3-connected, so again there is a vertex w such that $\{z, v, w\}$ separates G , and as before every vertex in $\{z, v, w\}$ has a neighbour in every component of $G - \{z, v, w\}$.

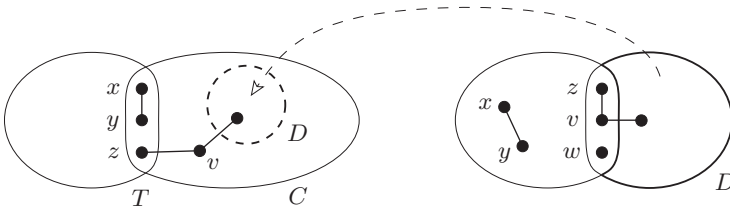


Fig. 3.2.1. Separating vertices in the proof of Lemma 3.2.4

As x and y are adjacent, $G - \{z, v, w\}$ has a component D such that $D \cap \{x, y\} = \emptyset$. Then every neighbour of v in D lies in C (since $v \in C$), so $D \cap C \neq \emptyset$ and hence $D \subsetneq C$ by the choice of D . This contradicts the choice of xy , z and C . $\square$

Theorem 3.2.5. (Tutte 1961)

A graph G is 3-connected if and only if there exists a sequence $G_0, \dots, G_n$ of graphs with the following two properties:

- (i) $G_0 = K^4$ and $G_n = G$;
- (ii) G_{i+1} has an edge xy such that $d(x), d(y) \geq 3$ and $G_i = G_{i+1}/xy$, for every $i < n$.

Moreover, the graphs in any such sequence are all 3-connected.

Proof. If G is 3-connected, then by Lemma 3.2.4 there is a sequence $G_n, \dots, G_0$ of 3-connected graphs satisfying (i) and (ii).

Conversely, and to show the final statement of the theorem, let $G_0, \dots, G_n$ be a sequence of graphs satisfying (i) and (ii); we show that if G_i is 3-connected then so is G_{i+1} , for every $i < n$. Suppose not, let S be a separator of at most 2 vertices in G_{i+1} , and let C_1, C_2 be two components of $G_{i+1} - S$. As x and y are adjacent, we may assume that $\{x, y\} \cap V(C_1) = \emptyset$ (Fig. 3.2.2). Then C_2 contains neither both vertices x, y nor a vertex $v \notin \{x, y\}$: otherwise v_{xy} or v would be separated from C_1 in G_i by at most two vertices, a contradiction. But now C_2 contains only one vertex: either x or y . This contradicts our assumption of $d(x), d(y) \geq 3$. $\square$

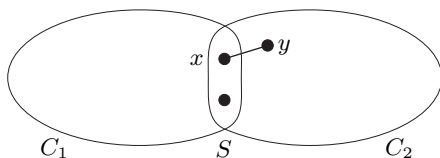


Fig. 3.2.2. The position of $xy \in G_{i+1}$ in the proof of Theorem 3.2.5

Like Theorem 3.2.3, Theorem 3.2.5 enables us to construct all 3-connected graphs inductively from K^4 , by simple local alterations and without ever leaving the class of 3-connected graphs. Given a 3-connected graph already constructed, pick any vertex v and split it into two adjacent vertices v', v'' ; then join these to all the former neighbours of v , each to at least two. This is the essential core of a result of Tutte known as his *wheel theorem*.⁴

For larger integers k it is no longer true that in any k -connected graph we can contract an edge so as to obtain another k -connected graph. However, for every k there is a constant n_k such that in every k -connected graph we can either delete or contract an edge so that the resulting graph has no separation of order less than k in which both sides have at least n_k vertices. See the notes.

Theorem 3.2.6. (Tutte 1963)

[4.5.2]

The cycle space of every 3-connected graph G is generated by its non-separating induced cycles.

Proof. Given a cycle C in G , let $b(C)$ denote the largest order of a component of $G - C$ if there is one, and put $b(C) := 0$ if $V(C) = V(G)$. Suppose the theorem is false, and consider a cycle C with $b(C)$ maximum that is not generated by non-separating induced cycles.

b

C

⁴ Graphs of the form $C^n * K^1$ are called *wheels*; thus, K^4 is the smallest wheel.

wheel

If $V(C) = V(G)$, then C is the sum of two cycles $C_1, C_2 \subseteq C + e$, where e a chord. As $b(C_1), b(C_2) > 0 = b(C)$, our choice of C implies that C_1 and C_2 are generated by non-separating induced cycles. But then so is their sum C , a contradiction.

Assume now that $G - C \neq \emptyset$, and let B be a component of $G - C$ of order $|B| = b(C)$. Suppose first that

$G - B$ contains a C -path $P = u \dots v$ such that each of the two $u-v$ paths on C has an inner vertex in $N(B)$. (*)

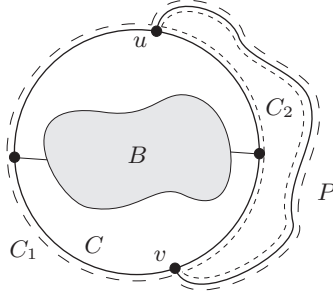


Fig. 3.2.3. C_1 and C_2 are drawn in broken lines

Then C is the sum of the two cycles $C_1, C_2 \subseteq C \cup P$ containing P , and for each of these C_i there is a component of $G - C_i$ that contains B properly (Fig. 3.2.3). Hence $b(C_i) > |B| = b(C)$, with a contradiction as earlier.

Suppose finally that (*) fails. Then every vertex of C sends an edge to B . (Indeed, if not then C contains an $N(B)$ -path $Q = x \dots y$ with $\dot{Q} \neq \emptyset$. As G is 3-connected, $C - Q \neq \emptyset$, and there is a $\dot{Q} - (C - Q)$ path in $G - \{x, y\}$. Such a path P would satisfy (*).) Since $V(C) = N(B)$, any chord of C would also be a path P as in (*), so C has no chord. Hence unless C itself is induced and non-separating, $G - C$ has a component $B' \neq B$. Let $P = u \dots v$ be a C -path through B' , and let Q be a C - P

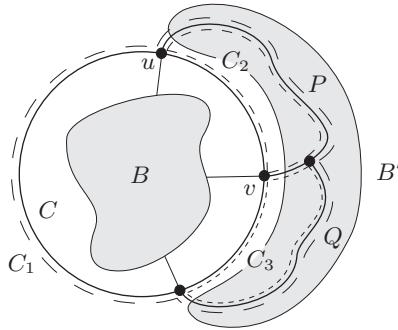


Fig. 3.2.4. Three cycles C_1, C_2, C_3 summing to C , and each missing a vertex of C that sends an edge to B

path in $G - \{u, v\}$. Note that Q too avoids B . Now $C \cup P \cup Q$ contains three cycles C_1, C_2, C_3 summing to C and each missing a vertex of C (Fig. 3.2.4). As every vertex of C sends an edge to B , we therefore have $b(C_i) > |B| = b(C)$ for every i , with the familiar contradiction. $\square$

3.3 Menger's theorem

The following theorem is one of the cornerstones of graph theory. It is another example of the particularly attractive type of theorem discussed in the notes for Chapter 2, a discussion which equally applies here:

Theorem 3.3.1. (Menger 1927)

Let $G = (V, E)$ be a graph and $A, B \subseteq V$. Then the minimum number of vertices separating A from B in G is equal to the maximum number of disjoint A – B paths in G .

[3.5.2]
[8.2.5]
[8.4.3]
[12.4.3]
[12.6.3]

We offer three proofs. Whenever G, A, B are given as in the theorem, we denote by $k(G, A, B)$ the minimum number of vertices separating A from B in G . Clearly, G cannot contain more than $k = k(G, A, B)$ disjoint A – B paths; our task will be to show that k such paths exist.

k

First proof. We apply induction on $\|G\|$. If G has no edge, then $|A \cap B| = k$ and we have k trivial A – B paths. So we assume that G has an edge $e = xy$. If G has no k disjoint A – B paths, then neither does G/e ; here, we count the contracted vertex v_e as an element of A (resp. B) in G/e if in G at least one of x, y lies in A (resp. B). By the induction hypothesis, G/e contains an A – B separator Y of fewer than k vertices. Among these must be the vertex v_e , since otherwise $Y \subseteq V$ would be an A – B separator in G . Then $X := (Y \setminus \{v_e\}) \cup \{x, y\}$ is an A – B separator in G of exactly k vertices.

We now consider the graph $G - e$. Since $x, y \in X$, every A – X separator in $G - e$ is also an A – B separator in G and hence contains at least k vertices. So by induction there are k disjoint A – X paths in $G - e$, and similarly there are k disjoint X – B paths in $G - e$. As X separates A from B , these two path systems do not meet outside X , and can thus be combined to k disjoint A – B paths. $\square$

Let $\mathcal{P}$ be a set of disjoint A – B paths, and let $\mathcal{Q}$ be another such set. We say that $\mathcal{Q}$ *exceeds* $\mathcal{P}$ if the set of vertices in A that lie on a path in $\mathcal{P}$ is a proper subset of the set of vertices in A that lie on a path in $\mathcal{Q}$, and likewise for B . Then, in particular, $|\mathcal{Q}| \geq |\mathcal{P}| + 1$.

exceeds

Second proof. We prove the following stronger statement: